

DIFFERENTIAL GEOMETRY

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ABSTRACT. Differential Geometry is regarded as a branch of mathematics that uses calculus and linear algebra to study the shapes, curves, and surfaces of smooth curved spaces. Unlike elementary geometry, which studies flat surfaces and rigid shapes, differential geometry focuses on local and global characteristics of curved spaces where geometric properties vary continuously from point to point. This framework serves as the mathematical language for modern theoretical physics.

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1. CHAPTER 0

1.1. Calculus in the Euclidean Space

This section is more or less an overture, intended to highlight some of the pre-requisite knowledge, language, and notation that would otherwise be helpful going forth through the subject.

1.2. The Euclidean Space

3-dimensional space is often regarded as the space of ordinary experience. It's the space we use to understand the world around us. To aid in developing of how we begin to describe the 3-dimensional space, we can often look at the corner of a table or a wall.

In the 3-dimensional space, rather than saying that 3 numbers describe the *position* of a point (or particle), we can take these 3 numbers to be a point itself.

Definition 1.1. E^3 is known as the **Euclidean 3-space**. It is the set of all ordered triples of real numbers.

$$\mathbf{p} = (p_1, p_2, p_3) \tag{1}$$

Eq. (1) denotes a point in E^3 . If $\mathbf{p} = (p_1, p_2, p_3)$ and $\mathbf{q} = (q_1, q_2, q_3)$ are both points in E^3 , then their sum and scalar product can be expressed (respectively) as:

$$\mathbf{p} + \mathbf{q} = (p_1 + q_1, p_2 + q_2, p_3 + q_3). \tag{2}$$

$$a\mathbf{p} = (ap_1, ap_2, ap_3). \quad (3)$$

Definition 1.2. Let $x, y, \&z$ be real-valued functions of E^3 such that for each point $\mathbf{p} = (p_1, p_2, p_3)$:

$$\begin{aligned} x(\mathbf{p}) &= p_1 \\ y(\mathbf{p}) &= p_2 \\ z(\mathbf{p}) &= p_3 \end{aligned}$$

They are known as the **natural coordinate functions** of E^3 . In index notation,

$$\begin{aligned} x_1 &= x \\ x_2 &= y \\ x_3 &= z \end{aligned}$$

This means that the value of a function x_i on a point p is the number p_i , revealing an important identity for each point in E^3 :

$$\mathbf{p} = (p_1, p_2, p_3) = (x_1(\mathbf{p}), x_2(\mathbf{p}), x_3(\mathbf{p})) \quad (4)$$

Definition 1.3. A real-valued function f on E^3 is **differentiable** if all partial derivatives of f , of all orders, exist and are continuous. This is also commonly known as **infinitely differentiable** or **of class C^∞** .

This states that a real-valued function is differentiable if all partial derivatives exist (meaning one can take the partial derivative with respect to each variable) and is continuous (smooth w/o abrupt changes). Similarly to points, real-valued functions are subject to operative manipulations:

$$(f + g)(\mathbf{p}) = f(\mathbf{p}) + g(\mathbf{p}), \quad (5)$$

$$(fg)(\mathbf{p}) = f(\mathbf{p})g(\mathbf{p}). \quad (6)$$

It is of importance to realise that differentiation is a *local* operation. This puts one's attention not in all space, but restricted to particular regions. In a more formalised manner:

Let f be a function.

Let D be the domain of f .

Let $D = P$, where P is an open set.

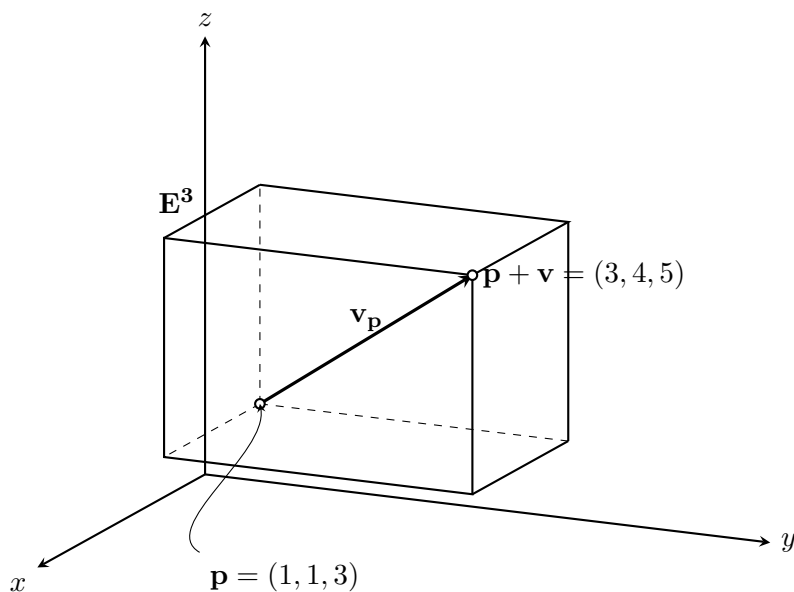
Let $P \subseteq \mathbf{E}^3$ such that if $p \in P$, then any $p_n \in P$ as well.

Note that the n -subscript means neighbour.

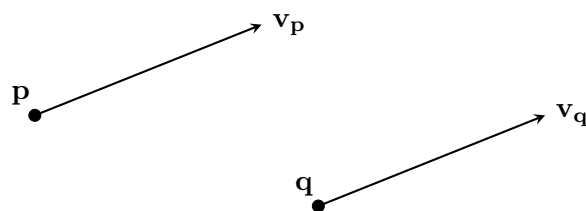
1.3. Tangent Vectors

Intuitively, we say that a vector in E^3 is simply described as an "arrow." "Arrows" in E^3 are more formally described by a starting point p and the change, or *vector* v , necessary to reach the end point, to which we denote as $p + v$.

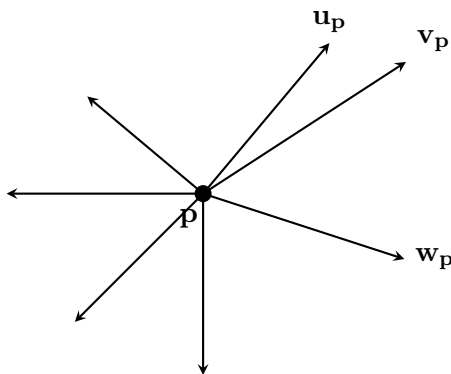
Definition 1.4. A **tangent vector** v_p to E^3 consists of two points of E^3 ; the vector part v and its point of application p .

Figure 1: The tangent vector $\mathbf{v}_p$ at point $\mathbf{p}$ in $\mathbf{E}^3$.

Tangent vectors $v_p = w_p$ are *equal* iff they share the same vector part $v = w$ and the same point of application $p = q$. Tangent vectors that share the same vector part but different points of application are said to be *parallel*.

Figure 2: Tangent vectors $\mathbf{v}_p$ and $\mathbf{v}_q$ at points $\mathbf{p}$ and $\mathbf{q}$.

Definition 1.5. Let p be a point of E^3 . The set $T_p(E^3)$ consisting of all tangent vectors that have p as the point of application is called the **tangent space** of E^3 at p

Figure 3: The tangent space $T_p(\mathbf{E}^3)$ consisting of all tangent vectors at point $\mathbf{p}$.

It is important to notice that at each point in space, E^3 has a different tangent space. This is because all tangent vectors share the same point of application. Once again, tangent vectors are subject to operative manipulations. These operations on each tangent space make it a vector space isomorphic to E^3 . This means that one can change the way a vector space *looks*, but fundamentally, it is still the same structure. A simplistic analogy is rotating a triangle.

A standard concept to note is that of a force field. For example, the gravitational field of Earth assigns to each point in space a force (vector) directed at the centre of the Earth.

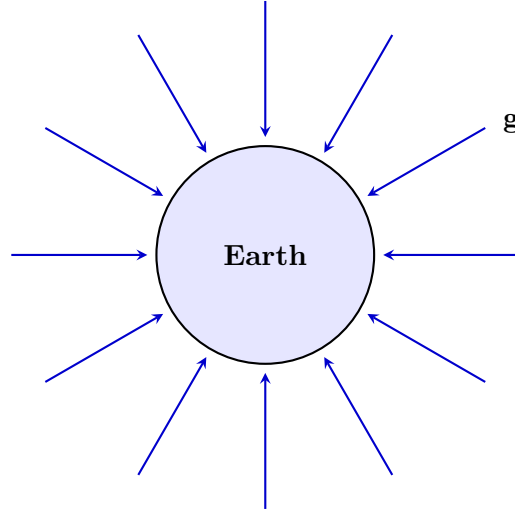


Figure 4: Gravitational force field vectors pointing inward toward the center of the Earth.

Definition 1.6. A **vector field** V on E^3 is a function that assigns to each point p in E^3 a tangent vector $V(p)$ to E^3 at point p . Simply, a vector field is a collection of arrows in E^3 . These too follow the previously defined operative manipulations.

Notably, there is a repetitive trend of being able to perform these "operative manipulations." This is what we call the **pointwise principle**; if a certain operation can be performed on the values of two functions at each point, then that operation can be extended to the functions themselves.

We next aim to describe what a vector field looks like.

Definition 1.7. Let U_1, U_2, U_3 be vector fields of E^3 such that:

$$U_1(\mathbf{p}) = (1, 0, 0)_p \quad (7)$$

$$U_2(\mathbf{p}) = (0, 1, 0)_p \quad (8)$$

$$U_3(\mathbf{p}) = (0, 0, 1)_p \quad (9)$$

for each point p of E^3 . U_1, U_2, U_3 are the **natural frame field** on E^3 .

Lemma 1.8. If V is a vector field on E^3 , then there are 3 uniquely determined real-valued functions v_1, v_2, v_3 on E^3 such that:

$$V = v_1 U_1 + v_2 U_2 + v_3 U_3 \quad (10)$$

v_1, v_2, v_3 are called the **Euclidean coordinate functions** of V .

Proof. We aim to show that $V = v_1U_1 + v_2U_2 + v_3U_3$. Recall the definition of a vector field (Def. 1.6). Also recall the identity expressed by Eq. (4).

$$\therefore V(\mathbf{p}) = (v_1(\mathbf{p}), v_2(\mathbf{p}), v_3(\mathbf{p}))_p$$

Multiplying each coordinate function by the natural frame fields (Eqs. (7,8,9)), we obtain:

$$v_1(\mathbf{p})U_1(\mathbf{p}) + v_2(\mathbf{p})U_2(\mathbf{p}) + v_3(\mathbf{p})U_3(\mathbf{p})$$

By the pointwise principle,

$$V = \sum v_i U_i$$

□

1.4. Directional Derivatives

Definition 1.9. Associated with each tangent vector v_p to E^3 is a straight line $t \rightarrow \mathbf{p} + t\mathbf{v}$. This essentially describes how the tangent vector changes over time. Specifically the vector point v .

Definition 1.10. If f is a differentiable function on E^3 , $f(t) \rightarrow f(\mathbf{p} + t\mathbf{v})$ is an ordinary differentiable function on a real line. The derivative of $f(t)|_{t=0}$ tells the initial rate of change of f as $\mathbf{p}$ moves in the $\mathbf{v}$ direction.

Definition 1.11. Let f be a differentiable real-valued function on E^3 , and let v_p be a vector tangent to E^3 . Then, the number

$$v_p[f] = \frac{d}{dt}(f(\mathbf{p} + t\mathbf{v}))|_{t=0}$$

is called the **derivative** with respect to v_p . $v_p[f]$ is called the **directional derivative**.

Lemma 1.12. For directional derivatives in general, let $v_p = (v_1, v_2, v_3)_p$ be a tangent vector to E^3 . Then,

$$\mathbf{v}_p[f] = \sum v_i \frac{\partial f}{\partial x_i}(\mathbf{p}).$$

Proof.

$$\mathbf{p} + t\mathbf{v} = (p_1 + tv_1, p_2 + tv_2, p_3 + tv_3).$$

Using the chain rule to compute the derivative at $t = 0$ of the function

$$f(\mathbf{p} + t\mathbf{v}) = f(p_1 + tv_1, p_2 + tv_2, p_3 + tv_3).$$

Since

$$\frac{d}{dt}(p_i + tv_i) = v_i,$$

we obtain

$$\mathbf{v}_p[f] = \frac{d}{dt}(f(\mathbf{p} + t\mathbf{v}))\Big|_{t=0} = \sum \frac{\partial f}{\partial x_i}(\mathbf{p})v_i.$$

□

Theorem 1.13. *Let f and g be functions on $\mathbf{E}^3$, $\mathbf{v}_p$ and $\mathbf{w}_p$ tangent vectors, a and b numbers. Then*

1. $(a\mathbf{v}_p + b\mathbf{w}_p)[f] = a\mathbf{v}_p[f] + b\mathbf{w}_p[f]$.
2. $\mathbf{v}_p[af + bg] = a\mathbf{v}_p[f] + b\mathbf{v}_p[g]$.
3. $\mathbf{v}_p[fg] = \mathbf{v}_p[f] \cdot g(\mathbf{p}) + f(\mathbf{p}) \cdot \mathbf{v}_p[g]$.

Proof. All three properties may be deduced easily from the preceding lemma.

Proof of (1)

Let $\mathbf{v} = (v_1, v_2, v_3)$ and $\mathbf{w} = (w_1, w_2, w_3)$. The tangent vector $a\mathbf{v}_p + b\mathbf{w}_p$ has components $av_i + bw_i$. By the lemma:

$$\begin{aligned} (a\mathbf{v}_p + b\mathbf{w}_p)[f] &= \sum (av_i + bw_i) \frac{\partial f}{\partial x_i}(\mathbf{p}) \\ &= \sum \left(av_i \frac{\partial f}{\partial x_i}(\mathbf{p}) + bw_i \frac{\partial f}{\partial x_i}(\mathbf{p}) \right) \\ &= a \sum v_i \frac{\partial f}{\partial x_i}(\mathbf{p}) + b \sum w_i \frac{\partial f}{\partial x_i}(\mathbf{p}) \\ &= a\mathbf{v}_p[f] + b\mathbf{w}_p[f]. \end{aligned}$$

Proof of (2)

By the lemma, if $\mathbf{v} = (v_1, v_2, v_3)$, then:

$$\mathbf{v}_p[af + bg] = \sum v_i \frac{\partial (af + bg)}{\partial x_i}(\mathbf{p}).$$

Using the linearity of partial derivatives, $\frac{\partial (af + bg)}{\partial x_i} = a \frac{\partial f}{\partial x_i} + b \frac{\partial g}{\partial x_i}$. Hence:

$$\begin{aligned} \mathbf{v}_p[af + bg] &= \sum v_i \left(a \frac{\partial f}{\partial x_i}(\mathbf{p}) + b \frac{\partial g}{\partial x_i}(\mathbf{p}) \right) \\ &= a \sum v_i \frac{\partial f}{\partial x_i}(\mathbf{p}) + b \sum v_i \frac{\partial g}{\partial x_i}(\mathbf{p}) \\ &= a\mathbf{v}_p[f] + b\mathbf{v}_p[g]. \end{aligned}$$

Proof of (3)

For example, we prove (3). By the lemma, if $\mathbf{v} = (v_1, v_2, v_3)$, then

$$\mathbf{v}_p[fg] = \sum v_i \frac{\partial (fg)}{\partial x_i}(\mathbf{p}).$$

But

$$\frac{\partial (fg)}{\partial x_i} = \frac{\partial f}{\partial x_i} \cdot g + f \cdot \frac{\partial g}{\partial x_i}.$$

Hence

$$\begin{aligned} \mathbf{v}_p[fg] &= \sum v_i \left(\frac{\partial f}{\partial x_i}(\mathbf{p}) \cdot g(\mathbf{p}) + f(\mathbf{p}) \cdot \frac{\partial g}{\partial x_i}(\mathbf{p}) \right) \\ &= \left(\sum v_i \frac{\partial f}{\partial x_i}(\mathbf{p}) \right) g(\mathbf{p}) + f(\mathbf{p}) \left(\sum v_i \frac{\partial g}{\partial x_i}(\mathbf{p}) \right) \\ &= \mathbf{v}_p[f] \cdot g(\mathbf{p}) + f(\mathbf{p}) \cdot \mathbf{v}_p[g]. \end{aligned}$$

□

From these proofs, one can gather that no matter what the form of differentiation is, it will always have suitable *linear* and *leibnizian* properties. We can also use the pointwise principle to define the operation of a vector field V on a function f . Similar to tangent vectors, the directional derivative of the vector field $V(p)$ is $V(p)[f]$.

Corollary 1.14. *If V and W are vector fields on $\mathbf{E}^3$, and f, g, h are real-valued functions, then*

1. $(fV + gW)[h] = fV[h] + gW[h]$.
2. $V[af + bg] = aV[f] + bV[g]$, for all real numbers a and b .
3. $V[fg] = V[f] \cdot g + fV[g]$.

Proof. Let $\mathbf{p} \in \mathbf{E}^3$ be an arbitrary point. By definition, a vector field acts on a function point-by-point according to the tangent vectors it assigns at each point (i.e., $V[h](\mathbf{p}) = V_{\mathbf{p}}[h]$). We evaluate each property at $\mathbf{p}$ using Theorem 3.3.

Proof of (1)

Applying the definition of the sum of vector fields at the point $\mathbf{p}$:

$$\begin{aligned} ((fV + gW)[h])(\mathbf{p}) &= (fV + gW)_{\mathbf{p}}[h] \\ &= (f(\mathbf{p})V_{\mathbf{p}} + g(\mathbf{p})W_{\mathbf{p}})[h]. \end{aligned}$$

Since $f(\mathbf{p})$ and $g(\mathbf{p})$ are real numbers, we apply property (1) of Theorem 3.3:

$$\begin{aligned} (f(\mathbf{p})V_{\mathbf{p}} + g(\mathbf{p})W_{\mathbf{p}})[h] &= f(\mathbf{p})V_{\mathbf{p}}[h] + g(\mathbf{p})W_{\mathbf{p}}[h] \\ &= f(\mathbf{p})(V[h](\mathbf{p})) + g(\mathbf{p})(W[h](\mathbf{p})) \\ &= (fV[h] + gW[h])(\mathbf{p}). \end{aligned}$$

Since this holds for all points $\mathbf{p}$, $(fV + gW)[h] = fV[h] + gW[h]$.

Proof of (2)

Evaluating the action of the tangent vector $V_{\mathbf{p}}$ on the linear combination $af + bg$:

$$(V[af + bg])(\mathbf{p}) = V_{\mathbf{p}}[af + bg].$$

Since a and b are real constants, we apply property (2) of Theorem 3.3:

$$\begin{aligned} V_{\mathbf{p}}[af + bg] &= aV_{\mathbf{p}}[f] + bV_{\mathbf{p}}[g] \\ &= a(V[f](\mathbf{p})) + b(V[g](\mathbf{p})) \\ &= (aV[f] + bV[g])(\mathbf{p}). \end{aligned}$$

Since this holds for all points $\mathbf{p}$, $V[af + bg] = aV[f] + bV[g]$.

Proof of (3)

Evaluating the action of the tangent vector $V_{\mathbf{p}}$ on the product function fg :

$$(V[fg])(\mathbf{p}) = V_{\mathbf{p}}[fg].$$

Applying property (3) of Theorem 3.3 (the Leibniz rule for tangent vectors):

$$\begin{aligned} V_{\mathbf{p}}[fg] &= V_{\mathbf{p}}[f] \cdot g(\mathbf{p}) + f(\mathbf{p}) \cdot V_{\mathbf{p}}[g] \\ &= (V[f](\mathbf{p})) \cdot g(\mathbf{p}) + f(\mathbf{p}) \cdot (V[g](\mathbf{p})) \\ &= (V[f] \cdot g + fV[g])(\mathbf{p}). \end{aligned}$$

Since this holds for all points $\mathbf{p}$, $V[fg] = V[f] \cdot g + fV[g]$. □

REFERENCES

- [1] Barrett O'Neill, *Elementary Differential Geometry*, Revised 2nd ed., Academic Press, 2006.